

FORM TP 2010021



TEST CODE **01238032**

JANUARY 2010

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

PHYSICS

Paper 03/2 – General Proficiency

Alternative to SBA

2 hours

**In addition to the 2 hours, candidates are allowed
a reading time of 10 minutes. Writing may begin
during the 10-minute period.**

READ THE FOLLOWING DIRECTIONS CAREFULLY.

1. You **MUST** use this answer booklet when responding to the questions. For each question, write your answer in the space provided and return the answer booklet at the end of the examination.
2. **ALL WORKING MUST BE SHOWN** in this booklet, since marks will be awarded for correct steps in calculations.
3. Attempt **ALL** questions.
4. The use of non-programmable calculators is allowed.
5. Mathematical tables are provided.

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1. A student plotted the graph shown in Figure 1 in an experiment to find the cooling curve of naphthalene.

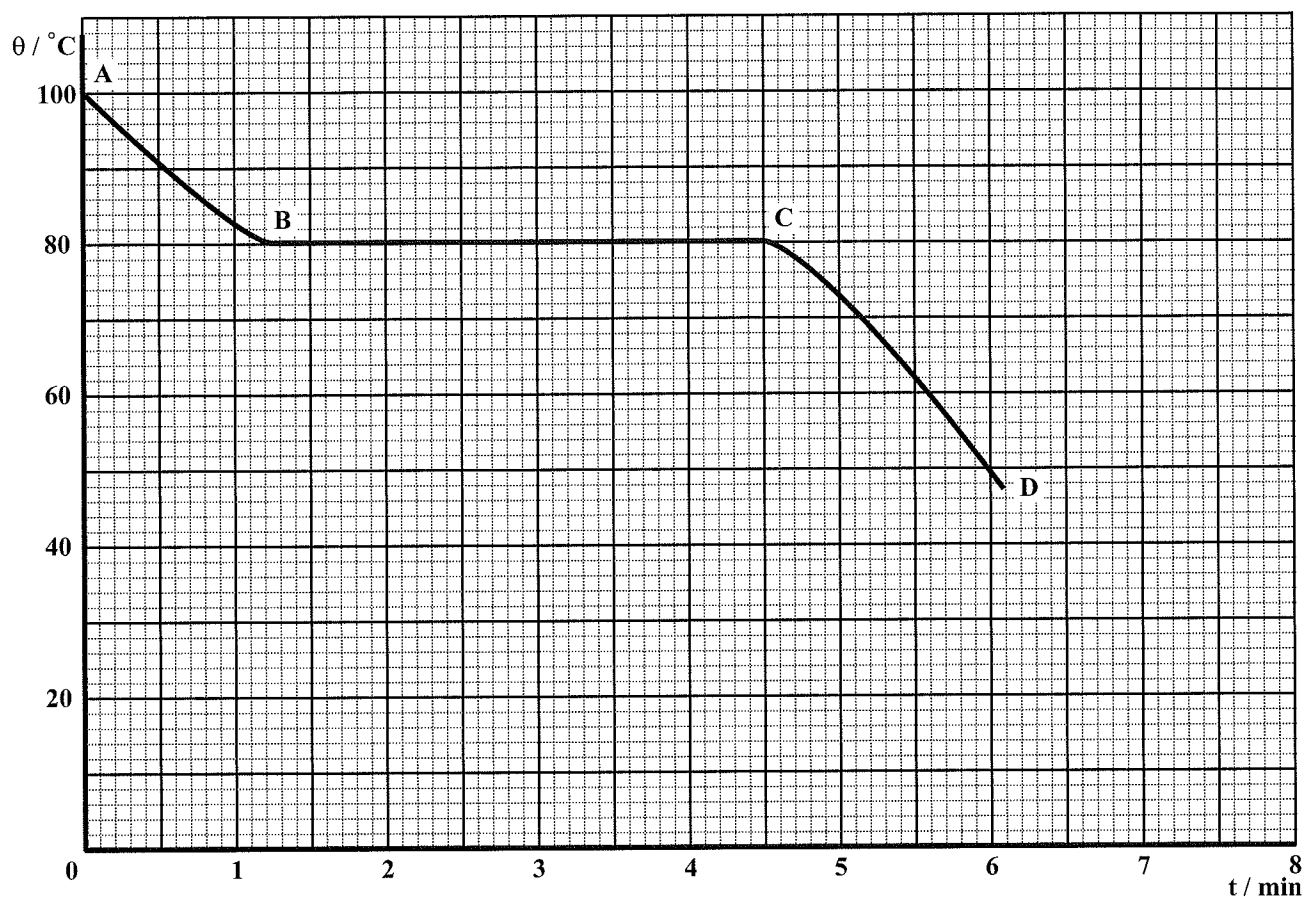


Figure 1. Cooling curve for naphthalene as plotted by the student

- (a) Complete Table 1 for values of θ and t using the graph shown in Figure 1.

TABLE 1

$\theta / ^{\circ}\text{C}$					80	80	80					50
t / min	0	0.5	1.0	1.4	2.0	3.0	4.0	4.7	5.0	5.5	5.6	6.0

(6 marks)

(b) Between which TWO points on the graph is naphthalene

(i) a solid? _____

(ii) a liquid? _____

(2 marks)

(c) What is the melting point of naphthalene in degrees Kelvin?

(3 marks)

(d) Write a brief description of a method the student could have used to conduct this experiment. Give ONE precaution to be taken and ONE source of error in outlining your method.

(6 marks)

Total 17 marks

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2. Two fourth form Physics students were asked to work together to investigate the relationship between extension, e , and the stretching force, F , for a spiral spring. Four sets of their results are shown in Figure 2.

(a) Use the diagrams shown in Figure 2 to complete Table 2.

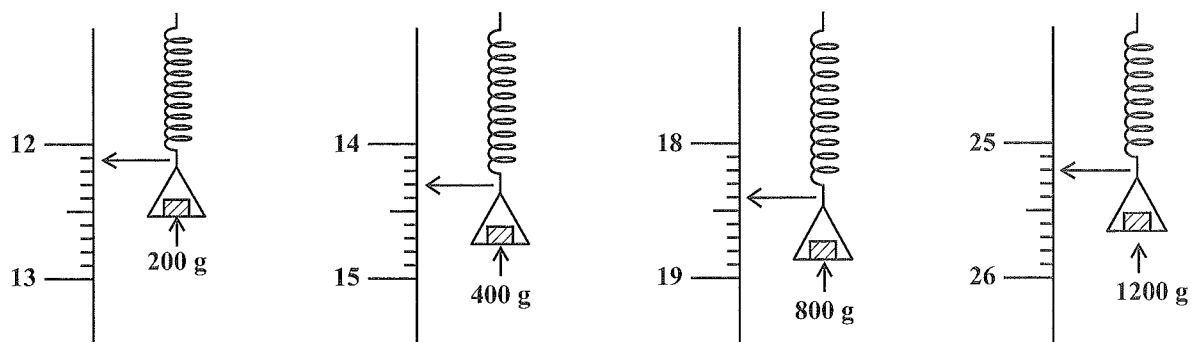


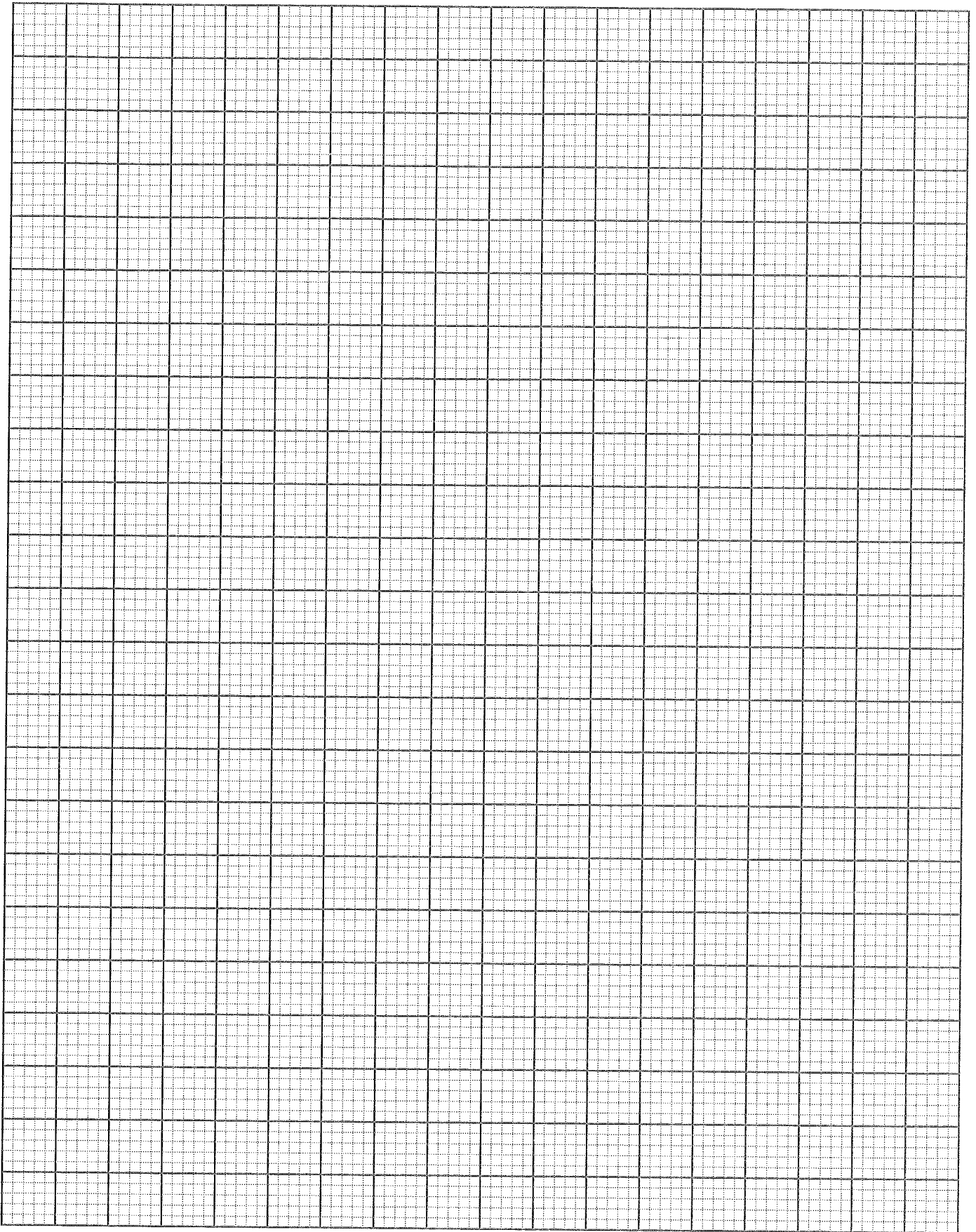
Figure 2. Diagram showing experiment

TABLE 2

Mass (m) /g	Stretching force (F) /N	Scale Reading /mm	Extension (e) /mm
0	0	10.0	0.0
200	2		
400	4		
600	6	16.3	6.3
800	8		
1000	10	21.4	11.4
1200	12		

(4 marks)

- (b) On the graph paper on page 5, plot a graph of extension, (e) against stretching force, (F).
(8 marks)
- (c) Identify the part of the graph where Hooke's law is NOT obeyed. Label the beginning and end of the part as A and B respectively.
(1 mark)



- (d) Calculate the gradient of the straight line portion of the graph.

(4 marks)

- (e) Use the graph to find the mass that would give an extension of 5.0 cm.
(Acceleration due to gravity = 10.0 m s^{-2}).

(2 marks)

- (f) Identify ONE possible source of error and ONE precaution to be taken in conducting this experiment.

(2 marks)

Total 21 marks

3. A student asserts that “the thicker a wire is, the greater will be its resistance to current”.

You are required to investigate whether or not this is true for Nichrome wire.

Your answer should include:

- (a) Apparatus list

(4 marks)

- (b) Procedure and diagram

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(3 marks)

(c) Use of observations

[illegible]

(3 marks)

Total 10 marks

END OF TEST